Causal Regime-Conditioned Trading Agent: Verifiable Regime-Aware Reinforcement Learning

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***Abstract*—Regime-aware reinforcement-learning (RL) trading agents are usually judged by back-tested returns, which cannot show whether the regime signal drives their decisions. We couple a Wasserstein regime-identity tracker with three interventions on the regime input, and we train a proximal policy optimisation (PPO) twin that differs only by lacking that input. On BTC/USDT four-hour bars the tracker lowers the mean template-matching cost by a factor of 4.9, but it issues no bull label after burn-in. On 11,430 out-of-fold bars ablation moves the conditioned position on 60.7% of bars and the agent de-risks in bear and crisis segments, yet only 0.1% of bars pass all three checks, against 33.5% for an archived moment-sorted labelling. The Sharpe ratios are -0.16 for the conditioned agent, -1.12 for the blind twin and 0.35 for buy-and-hold. Those four-hour numbers are the published single run. A five-seed repetition of the same recipe is reported beside them and does not replace them.**

The comparison in this paper is five seeds of timeframe-native twins on BTC/USDT and ETH/USDT, at 15 min and 1 h. A one-factor grid on the validation split did not meet the pre-registered bar: both twins with position standard deviation above 0.05, a share of bars with $|a| > 0.95$ below 0.80, and ablation of at least 0.10, on all four tapes. The baseline had the lowest mean deficit (about 0.985), so `configs/final.yaml` stayed the native recipe: raw reward, scale 100, entropy coefficient 0.01, uniform 2,000-bar windows, tracker v1, no perturbation and no faithfulness regularizer. Holdout and live were scored once and were not used to choose the recipe. The mean share of bars with $|a| > 0.95$ is 97–100% for every twin, and the mean holdout Sharpe ratio is negative for both twins. The conditioned agent beats the blind twin on a majority of seeds in two of eight windows only. On BTC 1 h holdout its mean return is $+2.8\%$ (standard deviation 38.2) against -26.1% (12.8), with beats share 0.80 and ablation 7.3% (13.6). On ETH 1 h live the means are $+0.4\%$ against -5.5%, with beats share 0.80. It does not win overall. The agents are not regularized, and both verifiers report small faithfulness rates.

Repeating the four-hour walk-forward study on seeds 42–46 does not turn that single run into a stable gain. Tracker v1's conditioned mean return is $+57.8\%$ only because seed 46 returns $+515\%$. The other four seeds are negative, and the mean Sharpe ratio is -0.27. Tracker v2's mean return is -83.6%, and the moment-sorted labelling's is -44.0%. Legacy faithfulness stays under 3% for both trackers and is 12.5% for moment-sorted names. On the native validation twins the raw Gaussian mean lies outside $[-1, 1]$ on 95.6–99.8% of bars, which is what the clip then executes as a rail. A tanh execution and a five-point discrete action were screened on the validation split only. Neither met the pre-registered bar on all four tapes, so holdout and live were not scored for those modes.

***Index Terms*—regime detection, hidden Markov model, Wasserstein distance, optimal transport, reinforcement learning, algorithmic trading, explainability, faithfulness, counterfactual testing**

I. Introduction

Cryptocurrency markets move through very different conditions within a few years. Bitcoin rose sharply in 2021, lost roughly three quarters of its value between late 2021 and late 2022, and then spent 2023–2026 alternating between recoveries and drawdowns. A trading policy fitted to such a history without any notion of market state tends to learn one averaged behaviour that suits none of these phases. A natural remedy is to detect the prevailing regime and hand it to the policy as an extra input. Ndoutoumou *et al.* combine a hidden Markov model with reinforcement-learning portfolio agents and report stronger risk-adjusted returns than the same agents without the regime signal [1]. RegimeFolio builds sector portfolios around detected regimes in non-stationary markets [2], and explainable classifiers have been used to allocate assets by market state [3].

What this work rarely checks is whether the regime input matters. A regime-conditioned agent can report that it "cut exposure because a crisis was detected" while its actual computation ignores the crisis probability, or responds to it in the wrong direction. An explanation that is easy to understand is not necessarily faithful to the computation that produced the decision [4]. Reviews of explainable machine learning find that explanations are often judged by whether users trust them, while standard tests of whether an explanation is true of the model are still missing [5]. Finance relies on much the same post-hoc tools, and their evaluation is uneven [6]. In high-stakes settings, the duty to explain a decision is not optional [7].

A second, quieter problem sits inside the regime detector. An HMM refitted on new data returns its states in an arbitrary order, so "state 2" may mean crisis in one window and calm in the next. Sorting states by their moments after every fit repairs the names locally, but it forces four names onto whatever four components the latest fit produces, even if one of them no longer resembles anything seen before. We instead keep persistent bull, bear, calm and crisis templates and match each new component to the nearest template by the closed-form 2-

Wasserstein distance, so a name survives a refit only when the component still resembles it.

This paper asks two linked questions. Does Wasserstein tracking give regime labels that keep their meaning across walk-forward refits? And when an RL agent is conditioned on those labels, does its behaviour depend on them in the way the labels' names imply? A four-hour walk-forward study on BTC/USDT answers both. The trading comparison for 15-min and 1-h candles is a five-seed study of timeframe-native twins on BTC/USDT and ETH/USDT, and that comparison is the result reported here. Our contributions are:

- A walk-forward implementation of Wasserstein template tracking for a four-state Gaussian HMM, with the closed-form diagonal W_2 distance, probability aggregation and template smoothing written out as equations that match the released code.
- A per-decision verification suite of three interventions on the regime input (ablation, counterfactual injection and a sensitivity sweep), validated on reference policies that are faithful, regime-ignoring and inverted by construction.
- A twin experiment in which regime-conditioned and regime-blind PPO agents share architecture, seed and training budget, evaluated walk-forward with bootstrapped confidence intervals and a per-regime breakdown.
- An empirical finding: replacing moment sorting with Wasserstein tracking leaves the regime input load-bearing but collapses the faithfulness rate from 33.5% to 0.1%. We trace this to template starvation and to counterfactual inputs that lie outside the training distribution.
- A live paper-trading console that runs both twins on closed Binance candles, and a five-seed study of timeframe-native twins on BTC/USDT and ETH/USDT at 15 min and 1 h. The validation grid did not meet the pre-registered bar, so the native recipe was frozen before holdout and live were scored. The conditioned twin beats the blind twin on a majority of seeds in two of eight windows only. Every twin spends 97–100% of bars at $|a| > 0.95$, and the mean holdout Sharpe ratio is negative for both twins.
- A five-seed repetition of the four-hour study, and a check of whether Gaussian clipping puts the native twins on the rails. The four-hour return mean is one seed. The raw policy mean lies outside $[-1, 1]$ on nearly every validation bar, and neither a tanh execution nor a five-point discrete action passed the pre-registered bar on every tape.

II. Related Work

A. Regime detection

Ndoutoumou, Yin and Cheng fit a Gaussian HMM and pass the inferred regime into reinforcement-learning agents on a Dow Jones portfolio. For several algorithms the hybrid agent improves the Sharpe and Sortino ratios relative to the same agent without the HMM [1]. The comparison is on back-tested returns. It does not test whether the learned policy actually depends on the regime input. Tsang and Chen detect regime change in the foreign-exchange market with directional-change indicators, and use the detector to chart the period around Brexit [8]. Zhang, Yi and Chen use explainable machine learning to classify regimes and to support asset allocation [3]. RegimeFolio is a regime-aware learning system for sectoral portfolio optimisation when markets are non-stationary [2]. Standard HMMs switch state too readily on noisy data; we suppress this with a simple minimum-dwell rule, and we keep regime names stable across refits by Wasserstein template matching, described in Section IV.

B. Reinforcement learning for portfolio optimisation

Deep reinforcement learning is now a standard way to treat portfolio choice as a sequential decision. Surveys cover the single-agent literature and the move toward multi-agent systems [9], [10]. Wang, Jin and Li cast asset-weight management as a Markov decision process and train a deep RL agent for diversification [11]. Tamuly, Bhutani and Sukriti apply deep RL to dynamic capital allocation [12]. Naik *et al.* use proximal policy optimisation and deep deterministic policy gradient for real-time rebalancing on Indian equities, with the Sharpe ratio as the risk-adjusted objective [13]. Huang and Jing adapt a single-stock portfolio with a deep deterministic policy gradient agent [14]. Gao and Tang evaluate advantage actor-critic and proximal policy optimisation on A-share portfolios drawn from CSI 300 constituents, under A-share trading constraints [15]. Shrirao *et al.* train portfolio policies intended to remain usable under adversarial market conditions [16]. Across this work, a higher Sharpe ratio is treated as evidence that the agent has adapted. None of it checks whether a named regime input is what the policy uses.

C. Explainability

Mamalakis *et al.* separate an explanation that is faithful to a deep network from one that is merely comprehensible [4]. Wiratsin and Ragkhitwetsagul review whether explainable-AI techniques improve human trust, and they note the lack of standard evaluation metrics [5]. Reinhardt and Buchholz argue that explainability in high-stakes domains is an obligation, not only a convenience [7]. Yeo *et al.* survey how post-hoc explanations are used in finance and how their limits affect decisions [6]. Our tests are different in kind: they intervene on one named input of a sequential policy rather than attributing a static prediction.

D. Causal inference in financial decisions

Qiu *et al.* build a causal-inference model for stock-trend forecasting, on the view that predictive accuracy alone does not identify which dependencies the forecast uses [17]. Katamaneni *et al.* study machine-learning causal inference for decisions that must be revised as the environment changes [18]. Both treat causality as a property of the market time series. The interventions in this paper are different. The policy is a known function, so setting its regime input to a chosen value

measures the effect of that input on the action, not a causal effect in the market.

III. Preliminaries

A. Gaussian hidden Markov model

Let $x_t \in \mathbb{R}^3$ be the regime features at bar t. A K-state Gaussian HMM has latent states $z_t \in \{1, \dots, K\}$, a transition matrix $A_{ij} = P(z_{t+1} = j \mid z_t = i)$, initial probabilities π_0 and emissions $x_t \mid z_t = k \sim \mathcal{N}(m_k, \Sigma_k)$. We use diagonal $\Sigma_k = \mathrm{diag}(v_k)$. Parameters are fitted by expectation–maximisation. The posterior $\gamma_t(k) = P(z_t = k \mid x_{1:n})$ comes from the forward–backward recursion and the most likely path from Viterbi decoding. The likelihood is invariant to permuting the state indices, so two fits on overlapping data can return the same regimes under different indices.

B. 2-Wasserstein distance between Gaussians

For probability measures μ, ν on $\mathbb{R}^d$ with finite second moments,

$$W_2^2(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu,\nu)} \int \|x - y\|^2 \, d\gamma(x, y), \tag{1}$$

where $\Gamma(\mu, \nu)$ is the set of couplings with marginals μ and ν. For Gaussians the infimum has the closed form

$$W_2^2\big(\mathcal{N}(m_1, \Sigma_1), \mathcal{N}(m_2, \Sigma_2)\big) = \|m_1 - m_2\|^2 + \mathrm{tr}\Big(\Sigma_1 + \Sigma_2 - 2\big(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2}\big)^{1/2}\Big). \tag{2}$$

With diagonal covariances $\Sigma_i = \mathrm{diag}(s_i^2)$ the matrices commute and the trace term reduces to $\|s_1 - s_2\|^2$, so

$$W_2^2 = \|m_1 - m_2\|^2 + \|s_1 - s_2\|^2. \tag{3}$$

It compares location and spread on the scale of the features and costs $O(d)$, so no numerical optimal-transport solver is needed.

C. Interventions on a policy input

A trained deterministic policy $a = \pi_\theta(o)$ is a known function of its observation. Write $o = (u, p)$, where p is the regime-probability block. We use $\pi_\theta(u, \mathrm{do}(p{=}p'))$ for the action obtained when p is set to p' and everything else is held fixed. Because π_θ is available in closed form, this intervention needs no causal identification assumptions. It measures the effect of the regime input on the policy's output, not any causal effect in the market. A regime signal is *load-bearing* if such interventions change the action. It is *faithful to its names* if the change has the sign that the names imply.

IV. Proposed Methodology

Fig. 1 shows the pipeline. All code is in the `crta` package of the project repository.

A. Data and causal features

Let C_t be the close of bar t. The features are the log return $r_t = \log(C_t/C_{t-1})$, the 20-bar rolling standard deviation σ_t of r, the trend $\tau_t = C_t/\mathrm{SMA}_{20}(C)_t - 1$, the 14-bar Wilder RSI scaled to $[0, 1]$, and the MACD histogram ($\mathrm{EMA}_{12} - \mathrm{EMA}_{26}$ minus its 9-bar signal) divided by C_t. Each feature at t uses rows $\le t$ only. A unit test checks that features computed on a truncated history equal the prefix of the full computation. The regime model uses $x_t = (r_t, \sigma_t, \tau_t)$. The agent sees all five as f_t, rescaled to unit order by a fixed map ϕ (r, σ, MACD $\times 100$; $\tau \times 10$; RSI$\mapsto 2\,\mathrm{RSI} - 1$).

B. Walk-forward HMM refits

The tracker makes a single forward pass over the series (Fig. 2). The first $B = 252$ bars (six weeks) form a burn-in. After that, refit j starts at $t_j = B + jS$ with stride $S = 42$ bars (one week). At each refit a four-state diagonal HMM is fitted by EM (200 iterations, seed 42) to the rows strictly before t_j, standardised with that window's mean $\bar{x}$ and standard deviation s. The model then decodes the window $W_j = \{t_j, \dots, t_j + S - 1\}$. Component parameters are mapped back to feature units so that different fits can be compared:

$$\mu_k = \hat{m}_k \odot s + \bar{x}, \qquad \sigma_k = \sqrt{\hat{v}_k} \odot s. \tag{4}$$

C. Template initialisation

The burn-in fit is the only place where names come from moments. The rule ℓ sorts components by mean return. Of the two lowest, the one with lower volatility is *bear* and the other is *crisis*. Of the two highest, the one with lower volatility is *calm* and the other is *bull*. This yields templates $(\bar{\mu}_g, \bar{\sigma}_g)$ for $g \in \mathcal{G} = \{\mathrm{bull}, \mathrm{bear}, \mathrm{calm}, \mathrm{crisis}\}$.

D. Wasserstein identity matching

At refit j the cost of naming component k as regime g is the diagonal Wasserstein distance (3):

$$D_{kg} = \|\mu_k - \bar{\mu}_g\|^2 + \|\sigma_k - \bar{\sigma}_g\|^2, \tag{5}$$

and every component takes its nearest template independently,

$$\pi_j(k) = \arg\min_{g \in \mathcal{G}} D_{kg}. \tag{6}$$

The matching need not be one-to-one: two components may share a template, and a template may receive none. The regime probabilities for $t \in W_j$ are the summed posteriors

$$p_t(g) = \sum_{k:\, \pi_j(k) = g} \gamma_t(k), \tag{7}$$

and an unmatched template has $p_t(g) = 0$.

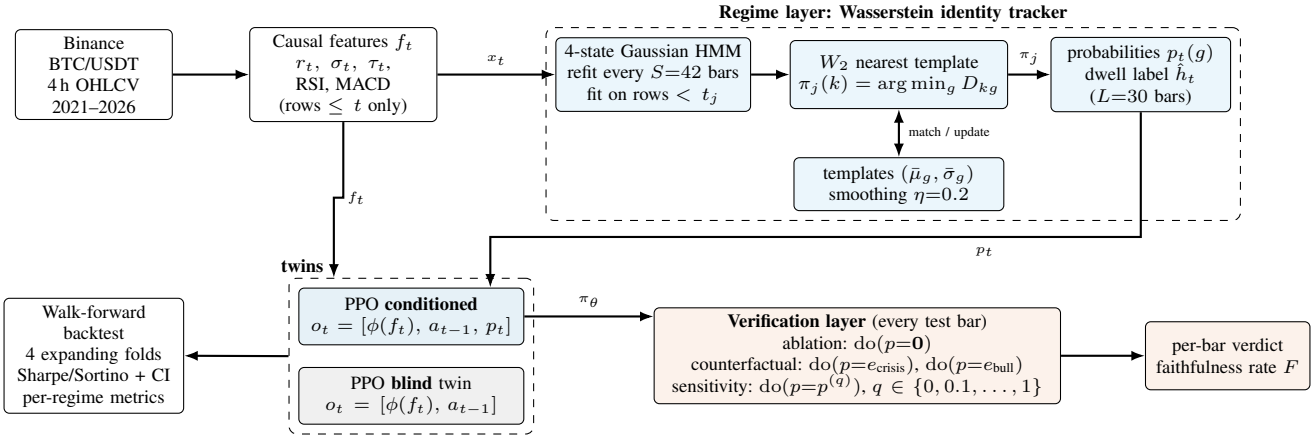


Fig. 1. System architecture. Features are causal. The regime layer refits the HMM on past rows only, names its components by Wasserstein distance to persistent templates, and passes the aggregated probabilities p_t to the conditioned agent. The blind twin differs only in lacking p_t. Every out-of-fold decision of the conditioned agent is audited by three interventions on p_t.

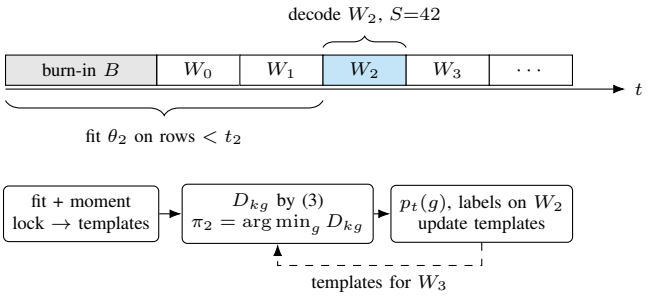


Fig. 2. Causal refit schedule. The burn-in fit names the templates once by moment sorting. Every later window is decoded by a model fitted only on earlier rows, and its components inherit names by Wasserstein matching. Labels already issued are never revised.

E. Template smoothing

After window W_j is labelled, every matched template moves a step $\eta = 0.2$ toward its members. Let $w_k = \sum_{t \in W_j} \gamma_t(k)$ and $\omega_k = w_k / \sum_{k' \in M_g} w_{k'}$ for the member set $M_g = \{k : \pi_j(k) = g\}$. Then

$$\bar{\mu}_g \leftarrow (1-\eta)\,\bar{\mu}_g + \eta \sum_{k \in M_g} \omega_k \mu_k, \tag{8}$$

and likewise for $\bar{\sigma}_g$. A template with $M_g = \emptyset$ is left unchanged. The update uses window W_j and affects only windows after it.

F. Hard labels and dwell constraint

The hard label $\hat{h}_t$ is the name of the Viterbi state, $\pi_j(\hat{z}_t)$. Runs shorter than $L = 30$ bars (five days) are merged into the preceding label, with the last issued label carried across window boundaries. Hard labels are used only for reporting and per-regime evaluation. The agent sees the probabilities p_t.

G. Twin agents

The trading environment is a Gymnasium episode over the training rows. At bar t the agent chooses a target position $a_t \in [-1, 1]$ and receives

$$R_t = a_t\, r_{t+1} - c\,|a_t - a_{t-1}|, \qquad c = 10^{-3}, \tag{9}$$

that is, the next log return less a 10-basis-point turnover cost. The conditioned agent observes $o_t = [\phi(f_t), a_{t-1}, p_t] \in \mathbb{R}^{10}$ and the blind twin observes $[\phi(f_t), a_{t-1}] \in \mathbb{R}^6$. Both are PPO agents from Stable-Baselines3 with identical settings (Table II). The only architectural difference is four extra input weights in the first layer, and that difference is the experiment.

H. Verification suite

For each out-of-fold bar we freeze the conditioned policy, the market features and the previous position, and intervene on p alone. Let $a_t = \pi_\theta(u_t, p_t)$ be the realised action, e_g the one-hot vector of regime g, and ρ a defensive rank with $\rho(\text{bull}) = 0 < \rho(\text{calm}) = 1 < \rho(\text{bear}) = 2 < \rho(\text{crisis}) = 3$.

Ablation. The regime input is load-bearing at t if

$$\Delta_t^{\text{ab}} = \left|a_t - \pi_\theta(u_t, \text{do}(p{=}\mathbf{0}))\right| \ \geq\ \delta_{\text{ab}} = 0.15. \tag{10}$$

Counterfactual. With $g_t = \arg\max_g p_t(g)$, forcing crisis must cut the position and forcing bull must raise it, each by at least $\delta_{\text{cf}} = 0.10$, whenever the forced regime differs in rank from g_t:

$$\rho(g_t) < 3 \Rightarrow a_t - \pi_\theta(u_t, \text{do}(p{=}e_{\text{crisis}})) \geq \delta_{\text{cf}}, \tag{11}$$

$$\rho(g_t) > 0 \Rightarrow \pi_\theta(u_t, \text{do}(p{=}e_{\text{bull}})) - a_t \geq \delta_{\text{cf}}. \tag{12}$$

Sensitivity. For $q \in \{0, 0.1, \ldots, 1\}$ we inject a crisis probability q and rescale the other entries:

$$p^{(q)} = (1-q)\,\frac{p_t^{\setminus c}}{\|p_t^{\setminus c}\|_1} + q\, e_{\text{crisis}}, \tag{13}$$

where $p_t^{\setminus c}$ is p_t with its crisis entry set to zero (uniform over the other three if that leaves nothing). The curve

$y_i = \pi_\theta(u_t, \mathrm{do}(p{=}p^{(q_i)}))$ passes if it rises at most once, $\sum_i \mathbb{1}[y_{i+1} > y_i + 10^{-8}] \le 1$.

A bar is *faithful* when all three checks pass. The faithfulness rate F is the share of faithful bars. Each bar costs 15 batched forward passes.

I. Live paper-trading console

The repository also includes an interactive console, built with Streamlit, in which both twins trade BTC/USDT with paper money as Binance candles close (Fig. 3). It never sends an order. The user chooses 15-min, 1-h or 4-h candles, as on a charting terminal, and every time stamp is shown in Indian Standard Time (IST, UTC+5:30).

Market feed. Candles come from Binance's public REST API through `ccxt`, which needs no key. A candle that opens at t_0 with width w is kept only once $t_0 + w \le t_{\text{now}}$, so a candle that is still forming never reaches an agent. The 4-h store used for training is updated incrementally. The console requests candles from the last stored open time, lets incoming rows replace overlapping ones, and writes the merged file to a temporary path before renaming it over the old one. If the request fails, the saved file is left untouched and the tape is marked stale. The 15-min and 1-h tapes keep their own caches of at most twice the display window (1,000 candles, or 500 for 4 h), so a faster timeframe cannot overwrite the training data. Every 20 s the console polls for newly closed candles and for the last traded price. The price is only displayed, and every decision uses closed candles.

Labels. When a tape is opened, one HMM is fitted on its first $\min(80, \max(20, \lfloor n/5 \rfloor))$ rows, its states are named by the moment rule, and the rest of the window is decoded in one pass. Each candle that closes later is decoded by a model fitted on all rows before it, with the last issued label carried across. This is faster than the full tracker but not identical to it. Section VI-E scores the live window with the tracker.

Paper books. Both agents start flat with unit wealth on the same tape. At each closed candle t each agent chooses a_t from its own observation. When candle $t{+}1$ closes, its wealth is multiplied by e^{R_t}, with R_t from (9) including the 10-bp turnover cost. The desk plots both wealth curves, each agent's latest fills and the regime vector fed to the conditioned agent. Sliders let the user override the four regime probabilities for the open candle (rescaled to sum to one), and optionally its features and held position. This applies $\mathrm{do}(p{=}p')$ to the conditioned agent alone and shows its bet next to the blind twin's, so the verifier's interventions can be explored by hand.

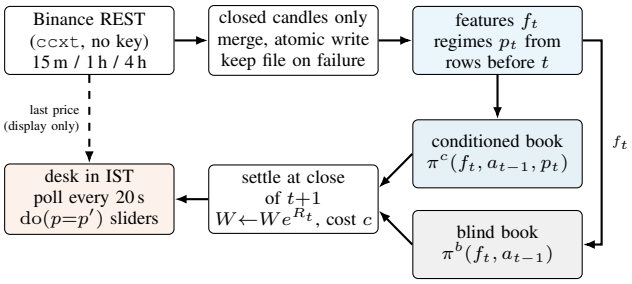


Fig. 3. Live paper-trading console. Only closed candles reach the agents. Both books trade the same tape and are settled with the training reward (9). The last traded price is shown but never used for a decision.

TABLE I
FOUR-HOUR WALK-FORWARD FOLDS AND PER-FOLD SHARPE RATIOS (ANNUALISED, OUT OF FOLD). THESE NUMBERS WERE NOT RECOMPUTED FOR THE FIVE-SEED RUN

Fold	Test period	Train	Cond.	Blind	B&H	Cond.†
0	07/2021–10/2022	1,080	−1.95	−1.84	−0.65	−2.32
1	10/2022–02/2024	3,937	0.24	−3.29	1.69	−0.53
2	02/2024–06/2025	6,794	0.53	0.66	1.10	2.24
3	06/2025–09/2026	9,651	0.50	−0.79	−0.39	1.93

†Archived run with per-fold moment-sorted labels (see Sec. V-C).

V. EXPERIMENTAL SETUP

A. Data and folds

We use Binance BTC/USDT four-hour candles from 1 Jan 2021 to 22 Sep 2026 (12,544 bars). Feature warm-up leaves 12,510 bars. Four expanding walk-forward folds start with 1,080 training bars, and each test block has about 2,857 bars (Table I). Each test block is scored by agents trained only on the bars before it. The concatenated test blocks give 11,430 out-of-fold bars from 5 Jul 2021 to 22 Sep 2026. Regime labels for every bar come from the single causal tracker pass described above.

B. Implementation details

Table II lists the settings. PPO uses the Stable-Baselines3 defaults for clipping (0.2) and epochs per update (10). Each agent trains for $\max(8192, 10\,N_{\text{train}})$ steps, which is ten passes over its training window: 10,800, 39,370, 67,940 and 96,510 steps for the four folds. HMMs are fitted with `hmmlearn`. If EM leaves a state with an all-zero transition row, that row is reset to uniform before decoding. The full pipeline (292 tracker refits, eight PPO trainings and the audit) ran in 6.6 minutes on a laptop CPU. All 32 unit tests of the repository pass. They include tests for feature leakage, the closed-form distance and label stability under a component swap. Others check that later bars never change a closed window and that a failed Binance request leaves the saved candles untouched. Further tests check that a refresh keeps older rows and replaces overlapping ones, and that the faster timeframes never write to the training cache. The last three cover the native-twin recipe: a windowed episode is one contiguous stretch of the series, no window replays the whole series, and the reward scale multiplies return and cost alike. Table II is the four-hour study. The native recipe in Section V-E uses an entropy coefficient of 0.01.

C. Baselines

Blind twin. The same PPO agent without p_t (Sec. IV-G). *Buy-and-hold.* A constant position $a_t = 1$. *Archived moment-sort run.* The repository keeps the outputs of the pipeline as it stood before the tracker was introduced. That version fitted one HMM per fold and named states with ℓ alone, and

TABLE II
IMPLEMENTATION SETTINGS OF THE FOUR-HOUR STUDY (FROM THE PROJECT CODE). THESE NUMBERS WERE NOT RECOMPUTED FOR THE FIVE-SEED RUN

Component	Setting
HMM	4 states, diagonal cov., EM 200 it., min. cov. 10^{-3}
Tracker	burn-in B=252, stride S=42, η=0.2, dwell L=30
Policy net	MLP $64-64$, $\tanh$, Gaussian action head
PPO	lr 3×10^{-4}, γ=0.99, λ_{GAE}=0.95, entropy 0
Rollout	n_{steps}=256, batch 64, seed 42, CPU
Cost	$c = 10^{-3}$ per unit turnover
Verifier	δ_{ab}=0.15, δ_{cf}=0.10, 11-point sweep, $\leq$1 uptick
Bootstrap	1,000 i.i.d. resamples, 95% percentile CI

everything else was identical. Its blind-twin and buy-and-hold results match ours exactly, so the labels are the only change. *Moment sorting on the same refits.* To isolate the identity rule, we also re-name every one of the 292 refits with ℓ instead of (6). This variant is used for label statistics only. *Reference policies for the verifier.* The rule $a = \text{clip}(p^\top w)$ with $w = (1, -0.4, 0.5, -1)$ over (bull, bear, calm, crisis) is faithful by construction. A constant $a = 0.5$ ignores the regime input, and $a = \text{clip}(-p^\top w)$ inverts it.

D. Metrics

Strategy log returns follow (9) with the realised positions. We report the annualised Sharpe ratio $\sqrt{2190}\,\bar{R}/\hat{s}_R$ and the Sortino ratio $\sqrt{2190}\,\bar{R}/\sqrt{\overline{\min(R,0)^2}}$, using 2,190 four-hour bars per year. Each comes with a 95% bootstrap interval. We also report maximum drawdown and total return. Every metric is additionally computed within each dwell-smoothed regime. For the tracker we report label shares, switches per year, run lengths, the mean matched W_2^2 cost, and an identity-jump statistic: the W_2^2 distance between the probability-weighted components that carry the same name at consecutive refits.

E. Timeframe-native twins

Data. The 15-min and 1-h tapes get twins of their own, trained on each timeframe's own candles. We add ETH/USDT as a second asset and treat it exactly like BTC/USDT, with its own tracker pass and its own twins. Binance candles from 1 Jan 2021 to the live snapshot give, after feature warm-up, 202,596 bars on 15 min and 50,627 bars on 1 h for each asset, about 16 and 4 times as many as the 4-h series.

Tracker. The tracker keeps the calendar spans of the 4-h study, a six-week burn-in, a weekly refit and a five-day dwell. In bars this is $(B, S, L) = (1{,}008, 168, 120)$ on 1 h and $(4{,}032, 672, 480)$ on 15 min. It runs once over each full series, so the live labels continue those of the training data.

Training recipe. One episode over the full 1-h training window, with the unscaled four-hour reward and no entropy bonus, sent both BTC twins towards a constant long position. The native twins are therefore trained on random windows. Each episode is a randomly placed stretch of 2,000 consecutive training bars, and the agent starts it flat. Training multiplies the reward (9) by 100 and discards that scale at evaluation. Each twin trains for 500,000 steps.

Frozen file. The validation grid in Section VI-E changes one factor at a time and scores only the validation split. No cell met the pre-registered bar, and the baseline cell had the lowest mean deficit (about 0.985). `configs/final.yaml` is the native recipe: raw reward, scale 100, entropy coefficient 0.01, uniform 2,000-bar windows, tracker v1, no regime perturbation and no faithfulness regularizer. The agents are not regularized. Tracker v2 was one cell of the grid and was not selected.

Splits. For validation, twins trained on bars before 22 Sep 2024 16:00 UTC are scored on the following year. For the holdout, twins retrained on bars before 22 Sep 2025 16:00 UTC are scored on the year up to the live window, and the same frozen weights are scored on the live window. Holdout and live were scored once and were not used to choose the recipe. The reported native runs use seeds 42 through 46.

Regime grouping. On fast candles the five-day dwell outlasts almost every run of the Viterbi path, so the hard label collapses to a single name over nearly the whole series: calm on BTC 1 h and ETH 15 min, bear on BTC 15 min, and bear on 94.7% of ETH 1-h bars. Per-regime results for the native twins are therefore grouped by the most likely regime $\arg\max_g p_t(g)$, the input the agent actually sees. Regime names are fixed per tape at burn-in, so the same name need not mean the same market state on different tapes.

F. Live evaluation protocol

Window. The last price used by the walk-forward study is the close of the 22 Sep 2026 12:00 UTC candle. The live window scores every candle that opened from 16:00 UTC (21:30 IST) that day until a snapshot taken on 13 Oct 2026 at 10:47 UTC. The last candle of each tape is dropped because its next return is not yet known. This leaves 1,994 decisions on 15-min candles and 497 on 1-h candles, over the same 21 calendar days. The candles are frozen with the study script, so the numbers below can be reproduced offline.

Agents. The native twins of Sec. V-E are trained on bars before 22 Sep 2025, for BTC and for ETH, at seeds 42 through 46. Weights stay fixed. Both twins start flat at the first live candle and carry their positions forward. Buy-and-hold enters at the same candle. The four-hour results in this paper are the out-of-fold study of Section VI-A.

Labels. Labels come from the causal tracker of Sec. IV, on each timeframe's own candles from 2021, with the calendar-scaled spans of Sec. V-E. The console fits a faster labelling when a tape is opened. Every live number in Section VI-E uses the tracker.

Why native twins. The same features are much smaller on a faster tape: the median σ_{20} is 0.14% on 15 min and 0.31% on 1 h, against 0.61% on 4 h and 1.04% over the four-hour training data. A four-hour conditioned agent transferred onto those tapes turned over 0.115 and 0.104 units per bar. The native twins are trained on the tape they trade.

Statistics. For the native twins we report the mean and the standard deviation across the five seeds. Sharpe ratios use 35,040 and 8,760 bars per year at 15 min and 1 h. "Beats" is

TABLE III
FOUR-HOUR REGIME LABELS AFTER BURN-IN (12,258 BARS): SHARE OF BARS, MEAN 4 H LOG RETURN AND MEAN σ_{20} PER LABEL. THESE NUMBERS WERE NOT RECOMPUTED FOR THE FIVE-SEED RUN

	W_2 tracker (ours)			Moment sort per refit		
Label	share	ret (bp)	σ (%)	share	ret (bp)	σ (%)
bull	0.0%	–	–	7.2%	−0.8	1.54
bear	7.7%	−7.7	1.42	34.2%	−3.1	0.93
calm	88.5%	+1.5	0.97	44.2%	+3.7	1.04
crisis	3.8%	−8.1	2.18	14.4%	−0.6	1.14
switches / yr		5.5			7.5	
median run (bars)		110			210	
mean matched W_2^2 (10^{-4})		1.88			9.13	

TABLE IV
OUT-OF-FOLD PERFORMANCE OF THE FOUR-HOUR STUDY, 11,430 BARS (95% BOOTSTRAP CI). THESE NUMBERS WERE NOT RECOMPUTED FOR THE FIVE-SEED RUN

System	Sharpe	Sortino	MDD	Return
Conditioned (W_2)	−0.16 [−0.98, 0.63]	−0.22 [−1.34, 0.91]	−77.0%	−29.3%
Blind twin	−1.12 [−1.96, −0.27]	−1.51 [−2.55, −0.37]	−94.4%	−93.0%
Buy-and-hold	0.35 [−0.51, 1.24]	0.50 [−0.71, 1.81]	−77.0%	+156.2%
Cond. (archived)	0.41 [−0.41, 1.20]	0.58 [−0.58, 1.77]	−79.8%	+139.2%

TABLE V
PER-REGIME SHARPE RATIO AND MEAN POSITION OF THE FOUR-HOUR STUDY (OUT OF FOLD, W_2 LABELS; NO BAR IS LABELLED BULL). THESE NUMBERS WERE NOT RECOMPUTED FOR THE FIVE-SEED RUN

		Sharpe			mean position	
Regime	n	Cond.	Blind	B&H	Cond.	Blind
bear	852	−0.19	−3.20	−1.94	−0.23	+0.17
calm	10,364	−0.17	−1.01	0.71	+0.41	+0.25
crisis	214	0.19	1.22	−2.23	−0.17	+0.24

the share of seeds whose conditioned log-return sum exceeds the blind twin. Ablation is the legacy pass rate. The four-hour Sharpe intervals in Tables I and IV use the i.i.d. bootstrap of Table II and were not recomputed here.

VI. RESULTS AND DISCUSSION

Tables III, IV and V keep the numbers of the four-hour study, and Figs. 4–9 were not redrawn. The five-seed comparison is Section VI-E.

A. Regime identity under refits

Fig. 4 shows the price with the tracker's labels, the bear and crisis probabilities, and both labellings. Table III compares them. Moment sorting assigns a different name than the W_2 rule to at least one component in 291 of the 292 refits, and to 49.6% of all components. Its labels are semantically unstable. Over the sample, the bars it calls *bull* have a negative mean return (-0.8 bp), and its *crisis* bars are only slightly more volatile than its *calm* bars (1.14% against 1.04%). The W_2 tracker keeps the ordering the names promise: crisis bars average 2.18% volatility and -8.1 bp per bar, against 0.97% and $+1.5$ bp for calm. It also switches less often (5.5 against 7.5 times per year). Its matched cost is 4.9 times lower (1.88 against 9.13×10^{-4}), and the identity jump of the crisis name falls by a factor of 7.4.

These gains have a price. Matching (6) is non-injective in 291 of the 292 refits. Fig. 5 shows why. The bull template inherits the burn-in's high-volatility, high-return component from early 2021. Later fits never produce a component near it, so it receives members in only 35 refits. Under (8) an unmatched template never moves, so after mid-2022 the bull template stays frozen. Calm absorbs the slack and covers 88.5% of bars, and no bar is labelled bull after burn-in. Fig. 6 shows a refit during the June 2022 crash. Components 0 and 2 both map to calm, while moment sorting would have called component 0 bull. A diagnostic variant (not part of the pipeline) replaces (6) with a one-to-one Hungarian assignment. It brings bull back to 5.1% of bars, with a mean return of $+11.9$ bp, but crisis then grows to 31.3% and its volatility falls to 1.15%. Neither rule alone keeps both coverage and sharp semantics.

B. Trading performance

Table IV and Fig. 7 summarise performance out of fold. With W_2 labels the conditioned agent has a Sharpe ratio of -0.16 and loses 29% over the test period. Its blind twin does much worse (-1.12, -93%), and buy-and-hold gains 156% with a Sharpe of 0.35. All bootstrap intervals for the conditioned agent overlap buy-and-hold, and only the blind twin's interval excludes zero. The archived run with moment-sorted labels reached 0.41, mostly through folds 2 and 3 (Table I). Its interval overlaps ours, so the regime-layer change cannot be said to have hurt returns significantly.

The per-regime view (Table V) shows where the gap comes from. In bear and crisis segments the conditioned agent holds a net short position on average (-0.23 and -0.17), while the blind twin stays long ($+0.17$ and $+0.24$). Its Sharpe ratio in bear segments, -0.19, is far better than the blind twin's -3.20 and buy-and-hold's -1.94. In calm segments it is long but trails buy-and-hold. Crisis segments cover only 214 test bars, about five weeks, so their annualised ratios are very noisy. This is consistent with an agent that uses its regime input, at least for regimes it saw in training.

C. Faithfulness

Fig. 8 reports the pass rate of each check. The reference policies behave as designed. The faithful rule passes on 97.9% of bars; it fails ablation only where its output is already within 0.15 of zero. The ignoring and inverted rules pass on no bar.

For the conditioned PPO agent with W_2 labels, ablation passes on 60.7% of bars, with a median $|\Delta^{\text{ab}}|$ of 0.42, so the regime input is clearly load-bearing. The counterfactual check, however, passes on only 0.9% of bars, and the combined faithfulness rate is 0.1% (11 of 11,430 bars). The archived

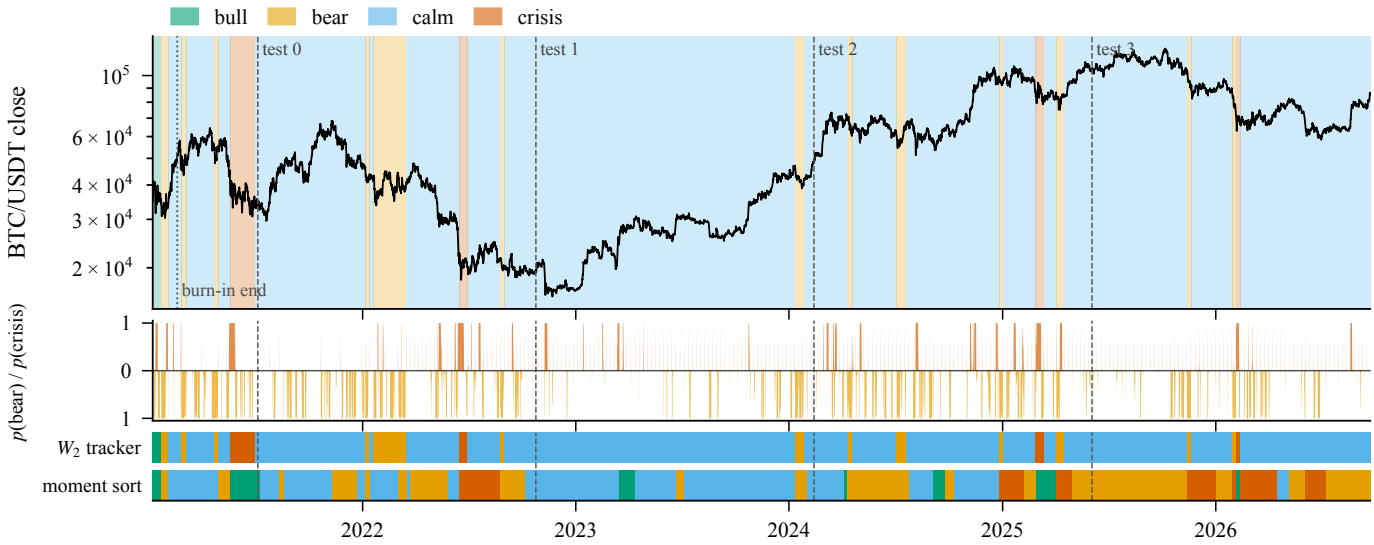


Fig. 4. BTC/USDT four-hour close (log scale) shaded by the dwell-smoothed W_2 tracker label (top). Tracker probabilities of crisis (up) and bear (down) (middle). Labels from the W_2 tracker and from moment sorting on the same 292 refits (bottom). Dashed lines mark the start of each test fold.

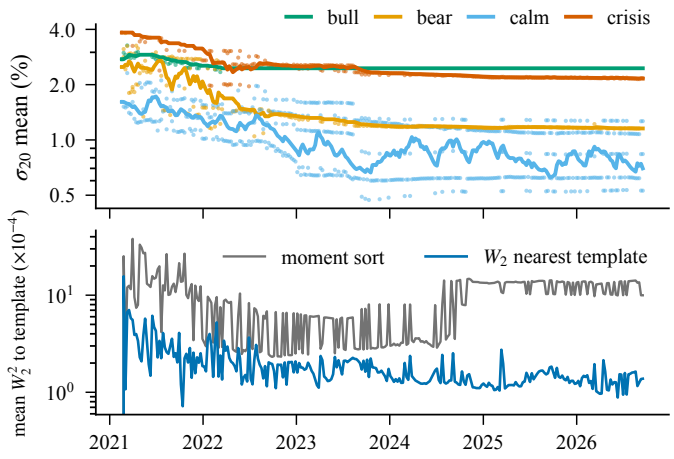


Fig. 5. Top: template volatility (lines) and fitted component volatilities coloured by their W_2 assignment (dots) at each refit. The bull template freezes once it stops receiving members. Bottom: mean squared W_2 distance between components and the template they are named after, for the W_2 rule and for moment sorting.

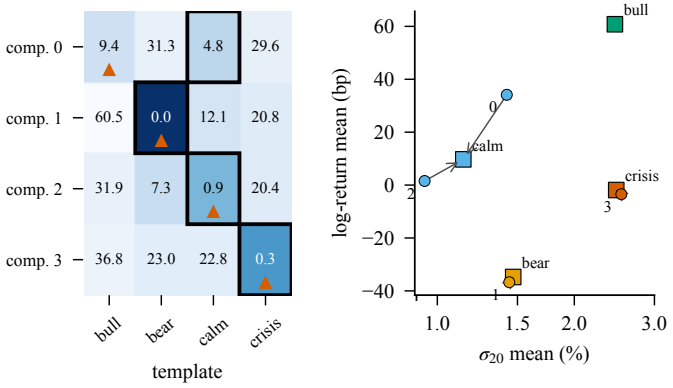


Fig. 6. Refit of 15 Jun 2022. Left: W_2^2 cost D_{kg} ($\times 10^{-4}$). Boxes mark the W_2 assignment and triangles the moment-sort assignment. Right: components (circles) and templates (squares) in mean-volatility and mean-return space, with arrows to the assigned template.

agent reached 33.5%. Fig. 9 explains the failure. Forcing *bear* sends the median position to -1, and the forced-bear position lies below the forced-calm position on 76.5% of bars. The agent has learned a defensive response, but it is keyed to bear. Forcing *crisis* gives a median of $+0.64$, which is defensive relative to calm on only 30.1% of bars. Forcing *bull* gives $+0.53$, below the realised median of $+0.74$. Taken separately, the two directions of the counterfactual check pass on 23.3% (crisis cuts the position) and 19.9% (bull raises it) of eligible bars, and their conjunction almost never holds. The crisis sweep in Fig. 9(a) tells the same story. For bear and crisis bars the mean position *rises* as crisis probability is injected.

This behaviour follows from the regime layer. The tracker's probabilities are almost one-hot (mean maximum 0.975), and crisis occupies only 3.8% of bars. One-hot bull never occurs in training after burn-in. The counterfactual inputs e_{bull} and e_{crisis} therefore lie far from, or entirely outside, the data the policy was fitted on. What the agent does there is an extrapolation, so the check fails without the agent being "miswired" on the inputs it actually meets. The archived labels kept all four regimes populated (bull 6.1%, crisis 12.4% of test bars), and their counterfactuals were in distribution. That, more than any quality of the agent, explains the archived 33.5%. Two further caveats concern the checks themselves. The counterfactual rank uses $\arg\max_g p_t(g)$, which agrees with the dwell-smoothed label on only 78.8% of bars. And the sensitivity check is weak, since a flat curve passes it. On 18 Jun 2022, near the crash low, $p_t(\text{crisis}) = 1$ and the agent is fully long. Every intervention returns $+1$, so ablation and

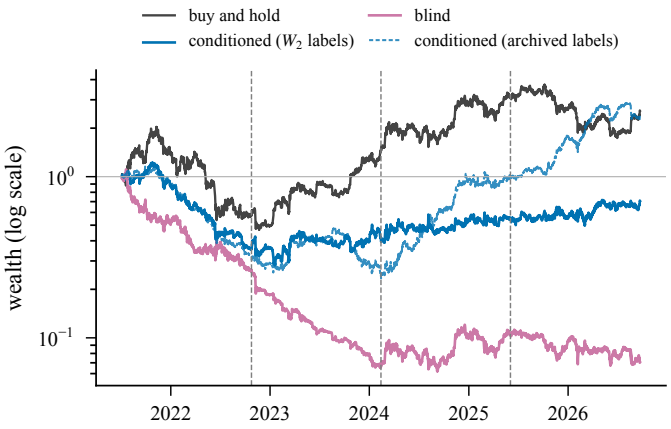


Fig. 7. Out-of-fold wealth (log scale). Dashed vertical lines separate test folds. The dashed blue curve is the archived conditioned agent trained on moment-sorted labels.

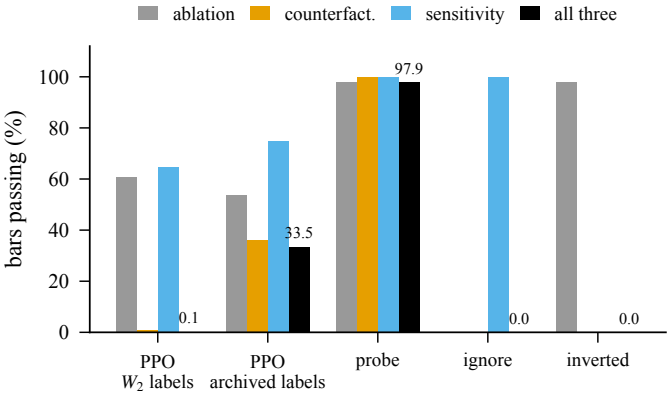


Fig. 8. Share of out-of-fold bars passing each check, for the PPO agent trained on W_2 labels, the archived PPO agent trained on moment-sorted labels, and three reference policies scored on the W_2 probabilities. Numbers give the share passing all three checks.

counterfactual fail while sensitivity passes. The ignoring policy passes sensitivity on every bar for the same reason.

The lesson is methodological. Swapping one well-motivated regime component for another, with architecture, data and training unchanged, improved label semantics and reversed the faithfulness verdict. A verdict cannot be carried over to a modified pipeline.

D. Five-seed repetition of the four-hour study

Tables IV and V and Figs. 7–9 stay the published single run. They were not overwritten. The same walk-forward recipe was repeated for seeds 42–46 on the frozen candles through 22 Sep 2026 12:00 UTC. Tracker v1, tracker v2 and moment-sorted names were each trained from scratch. Moment-sorted names are assigned on the same causal refits as tracker v1. The budget stayed $\max(8192, 10N_{\text{train}})$, with raw reward, scale 1 and entropy coefficient 0. Each labelling was scored by the legacy verifier and by verifier v2. The return does not depend on the verifier.

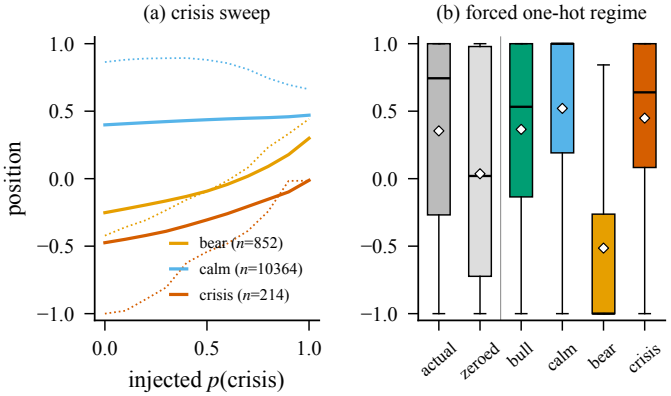


Fig. 9. Interventions on the conditioned agent (W_2 labels). (a) Mean (solid) and median (dotted) position against injected crisis probability, grouped by the bar's label. (b) Distribution of the realised position, the position with the regime input zeroed, and positions under one-hot regime inputs (boxes: interquartile range; diamonds: means).

Table VI is the mean and the standard deviation of those five seeds. The tracker-v1 conditioned mean return is $+57.8\%$ (standard deviation 257). Four of the five seeds lose money, -37.5%, -96.3%, -61.6% and -30.8%. Seed 46 returns $+515\%$. Of its four test folds, fold 2 alone returns $+870\%$, against $+111\%$ for buy-and-hold on that same fold. The other three folds return $+23\%$, -56% and $+16\%$. The mean Sharpe ratio stays negative, -0.27 (0.78). One wealth path dominates the average of the returns and does not dominate the average of the Sharpe ratios. Seed 42 of this repetition returns -37.5% with Sharpe -0.22, next to the published -29.3% and -0.16. The frozen file differs from the published run by one restored close. A single seed moves when that close moves, which is the same fact the five-seed spread is showing.

Tracker v2 loses on every seed. Its conditioned mean return is -83.6% (17.6) and its mean Sharpe ratio is -0.96 (0.47). Moment-sorted names lose on four seeds and gain $+116\%$ on seed 46, so the mean return is -44.0% (92.3). The blind twin does not see the labels, so its series is the same for every labelling: mean return -87.1% (7.6), Sharpe -0.93 (0.36). Buy-and-hold remains $+156.2\%$ with Sharpe 0.35.

The mean share of bars with $|a| > 0.95$ is 0.71–0.82. That is below the native twins and still a policy that lives on the rails. Ablation passes on 28.9% (24.8) of bars for tracker v1, 27.7% (12.9) for tracker v2 and 42.9% (13.8) for moment-sorted names. Faithfulness does not follow the ablation rate. The legacy all-check rate is 0.7% (0.8) for tracker v1, 2.4% (2.9) for tracker v2 and 12.5% (9.6) for moment-sorted names. Verifier v2 sits at or below the legacy rate in each case: 0.6%, 2.0% and 11.5%.

E. Five-seed native twins

The native recipe is the control cell of a one-factor grid. Each cell trains both twins for 500,000 steps on the validation split only (window 2,000, reward scale 100, two threads, seed 42), on BTC and ETH at 1 h and 15 min. The factors, one at a time, are the reward (raw, excess over buy-and-hold,

TABLE VI

FIVE-SEED REPETITION OF THE FOUR-HOUR WALK-FORWARD STUDY. ENTRIES ARE THE MEAN ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THOSE FIVE NUMBERS IN PARENTHESES. RETURN IS THE OUT-OF-FOLD TOTAL RETURN IN PERCENT. ABLATION AND FAITHFULNESS ARE SHARES OF OUT-OF-FOLD BARS, IN PERCENT. FAITHFULNESS IS THE SHARE PASSING ALL THREE CHECKS. THE BLIND TWIN DOES NOT DEPEND ON THE LABELLING. BUY-AND-HOLD IS $+156.2\%$ WITH SHARPE 0.35 ON EVERY SEED. THESE NUMBERS DO NOT REPLACE TABLE IV.

	conditioned		conditioned			faithfulness	
Labelling	return (%)	Sharpe	pos. s.d.	$\lvert a\rvert > 0.95$	ablation (%)	legacy	v2
Tracker v1	$+57.8$ (257)	-0.27 (0.78)	0.88 (0.09)	0.79 (0.11)	28.9 (24.8)	0.7 (0.8)	0.6 (0.8)
Tracker v2	-83.6 (17.6)	-0.96 (0.47)	0.89 (0.05)	0.79 (0.09)	27.7 (12.9)	2.4 (2.9)	2.0 (2.6)
Moment-sorted	-44.0 (92.3)	-0.95 (0.96)	0.82 (0.06)	0.71 (0.04)	42.9 (13.8)	12.5 (9.6)	11.5 (9.7)
Blind twin	-87.1 (7.6)	-0.93 (0.36)	0.90 (0.04)	0.82 (0.10)	–	–	–

differential Sharpe), the window sampler (uniform, regime-balanced, return-balanced), the entropy coefficient (0.01 or 0.02), a per-episode perturbation of the regime probabilities (0, 0.05 or 0.1), and the tracker (v1 or v2). Returns are not a selection input. A tape would pass only if both twins had position standard deviation above 0.05, both spent less than 80% of bars at $|a| > 0.95$, and ablation moved the conditioned agent on at least 10% of bars. The score of a cell is the mean, over the four tapes, of the summed relative shortfalls from those three bars.

No cell passed. Every validation twin still spent more than 90% of bars at $|a| > 0.95$. The baseline had the lowest mean deficit, about 0.985. The next cell, perturbation probability 0.1, was 1.33, and excess-over-buy-and-hold was the worst at 2.65: on BTC 1 h both of those twins finished with position standard deviation 0. Differential Sharpe did raise the standard deviation, often above 0.9, but the mass stayed on the rails and the BTC 15-min blind twin was constant. Return-balanced sampling produced an ablation rate of 73% on BTC 1 h, with standard deviations 0.56 and 0.84 and extreme shares of 0.98 and 1.00. No factor reduced the baseline deficit, so the pre-registered stack of helpful factors was empty and no combination was trained. `configs/final.yaml` stayed the native recipe: raw reward, scale 100, entropy coefficient 0.01, uniform 2,000-bar windows, tracker v1, no perturbation and no faithfulness regularizer. The agents below are not regularized.

The same frozen file was then trained five times, seeds 42 through 46, on bars before 22 Sep 2025 16:00 UTC. The holdout year and the 21-day live window are the splits in Secs. V-E and V-F. Holdout and live were scored once each and were not used to change the file. Table VII is the comparison. Entries are means across the five seeds. The table reports net return, position standard deviation, the share of seeds on which the conditioned log-return sum exceeds the blind twin, and the legacy ablation rate.

Buy-and-hold lost 23.3% and 23.7% on the BTC holdout tapes and 34.1% and 34.7% on the ETH holdout tapes (Table VIII). On BTC 1 h holdout the conditioned mean return is $+2.8\%$ (standard deviation 38.2) against -26.1% (12.8) for the blind twin. The beats share is 0.80 and the ablation rate is 7.3%. The mean Sharpe ratio on that window is still negative for both twins. On BTC 1 h live the conditioned mean return is 0.0% against $+1.8\%$ for the blind twin, and the beats share is 0.40.

On BTC 15 min holdout the conditioned mean return is -27.1% against -25.7% for the blind twin, the beats share is 0.20 and ablation is 0.1%. On BTC 15 min live the means are -1.5% against -2.1%, the beats share is 0.20, and the blind position standard deviation is 0 on all five seeds.

On ETH 1 h holdout the conditioned mean return is -12.8% (56.4) against -19.0% (39.6) for the blind twin. The beats share is only 0.20: the mean is one large path. On ETH 1 h live the conditioned mean return is $+0.4\%$ against -5.5%, the beats share is 0.80 and ablation is 29.5%.

On ETH 15 min holdout the conditioned mean return is -38.4% against -7.7% for the blind twin, and the beats share is 0.40. On ETH 15 min live the means are -5.4% against $+0.6\%$, and the beats share is 0.40.

The conditioned twin beats the blind twin on a majority of seeds in two of eight windows only: BTC 1 h holdout and ETH 1 h live. It does not win overall.

Table VIII reports the Sharpe ratios and the share of bars with $|a| > 0.95$. Every holdout Sharpe ratio is negative for both twins. Every twin spends 97–100% of bars at $|a| > 0.95$: the mean share on each row runs from 0.971 to 1.000. Where Table VII shows a large position standard deviation, the seeds disagree, and the positions that occur still lie at the rails of $[-1, 1]$. The ETH 1 h blind twin is flat on both windows (position standard deviation 0.02). The BTC 15-min live blind twin is flat: its position standard deviation is 0 on all five seeds. The median absolute ablation effect is 0 on every tape except ETH 1 h live, where the mean of the five medians is 0.27 and their standard deviation is 0.60. Ablation rates of 25.6% and 29.5% on ETH 1 h leave the median bar unmoved on the holdout; on the live window the median moves on one seed.

Figs. 10 and 11 draw each holdout and live path. Faint lines are seeds and the heavy line is the mean.

Fig. 12 and Table IX compare the legacy one-hot verifier with verifier v2. The v2 rate is at or below the legacy rate on every row, and both rates are small. The largest legacy mean is 0.046, on ETH 1 h holdout, and the v2 mean there is 0.036. The bull regime was testable on these tapes: no bar was marked untestable. The agents were not trained with the faithfulness penalty, so the check is an independent audit.

TABLE VII

FIVE-SEED NATIVE TWINS AFTER THE FROZEN RECIPE. ENTRIES ARE THE MEAN ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THOSE FIVE NUMBERS IN PARENTHESES. RETURN IS NET OF TURNOVER. "BEATS" IS THE SHARE OF SEEDS WHOSE CONDITIONED LOG-RETURN SUM EXCEEDS THE BLIND TWIN. ABLATION IS THE LEGACY PASS RATE. THE AGENTS ARE NOT REGULARIZED. THIS IS THE MAIN COMPARISON. THE MEAN SHARE OF BARS WITH $|a| > 0.95$ IS 97–100% FOR EVERY TWIN, AND EVERY HOLDOUT SHARPE RATIO IS NEGATIVE (TABLE VIII).

		conditioned		blind			
Tape	Window	return (%)	pos. s.d.	return (%)	pos. s.d.	beats	ablation (%)
BTC 1 h	holdout	2.8 (38.2)	0.55 (0.36)	−26.1 (12.8)	0.44 (0.45)	0.80	7.3 (13.6)
BTC 1 h	live	0.0 (5.6)	0.30 (0.44)	1.8 (4.8)	0.44 (0.45)	0.40	7.5 (14.6)
BTC 15 min	holdout	−27.1 (4.9)	0.09 (0.19)	−25.7 (2.8)	0.15 (0.33)	0.20	0.1 (0.3)
BTC 15 min	live	−1.5 (4.8)	0.17 (0.37)	−2.1 (3.1)	0.00 (0.00)	0.20	0.1 (0.2)
ETH 1 h	holdout	−12.8 (56.4)	0.74 (0.22)	−19.0 (39.6)	0.02 (0.04)	0.20	25.6 (20.0)
ETH 1 h	live	0.4 (9.3)	0.53 (0.14)	−5.5 (8.3)	0.02 (0.05)	0.80	29.5 (30.2)
ETH 15 min	holdout	−38.4 (8.8)	0.36 (0.29)	−7.7 (44.5)	0.37 (0.37)	0.40	0.5 (0.5)
ETH 15 min	live	−5.4 (3.2)	0.33 (0.43)	0.6 (8.9)	0.32 (0.44)	0.40	0.2 (0.2)

TABLE VIII

SHARPE RATIO, SHARE OF BARS WITH $|a| > 0.95$, AND BUY-AND-HOLD RETURN FOR THE FIVE-SEED NATIVE TWINS. SHARPE AND THE EXTREME SHARE ARE THE MEAN ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THOSE FIVE NUMBERS IN PARENTHESES. BUY-AND-HOLD HAS A FIXED POSITION, SO ITS RETURN DOES NOT VARY WITH THE SEED. EVERY HOLDOUT SHARPE RATIO IS NEGATIVE. THE EXTREME-SHARE MEANS RUN FROM 0.971 TO 1.000.

		Sharpe		$\lvert a\rvert > 0.95$		
Tape	Window	Cond.	Blind	Cond.	Blind	B&H (%)
BTC 1 h	holdout	−0.06 (0.86)	−0.72 (0.44)	0.988 (0.015)	0.984 (0.023)	−23.3
BTC 1 h	live	−0.07 (3.12)	0.94 (2.66)	0.995 (0.006)	0.985 (0.026)	−3.3
BTC 15 min	holdout	−0.71 (0.15)	−0.66 (0.09)	0.996 (0.008)	1.000 (0.000)	−23.7
BTC 15 min	live	−0.90 (2.68)	−1.22 (1.73)	0.996 (0.006)	1.000 (0.000)	−3.5
ETH 1 h	holdout	−0.52 (1.11)	−0.47 (0.65)	0.971 (0.028)	0.997 (0.007)	−34.1
ETH 1 h	live	−0.03 (4.14)	−2.68 (3.71)	0.982 (0.018)	0.993 (0.016)	−8.5
ETH 15 min	holdout	−0.79 (0.23)	−0.30 (0.84)	0.997 (0.003)	0.997 (0.006)	−34.7
ETH 15 min	live	−2.55 (1.54)	0.11 (4.08)	0.999 (0.002)	0.997 (0.005)	−8.8

TABLE IX

MEAN SHARE OF BARS PASSING ALL THREE FAITHFULNESS CHECKS, ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THE FIVE RATES IN PARENTHESES. LEGACY IS THE ONE-HOT VERIFIER. VERIFIER V2 USES TRAINING-SET PROBABILITY VECTORS AND REJECTS A FLAT SENSITIVITY CURVE. THE AGENTS ARE NOT REGULARIZED. THESE ARE THE RATES IN FIG. 12.

Tape	Window	Legacy	Verifier v2
BTC 1 h	holdout	0.006 (0.008)	0.005 (0.007)
BTC 1 h	live	0.001 (0.002)	0.001 (0.002)
BTC 15 min	holdout	0.001 (0.003)	0.001 (0.003)
BTC 15 min	live	0.001 (0.002)	0.001 (0.002)
ETH 1 h	holdout	0.046 (0.070)	0.036 (0.067)
ETH 1 h	live	0.035 (0.041)	0.021 (0.041)
ETH 15 min	holdout	0.003 (0.006)	0.003 (0.006)
ETH 15 min	live	0.001 (0.002)	0.001 (0.002)

Table X breaks the same seeds out by $\arg\max_g p_t(g)$. Several cells have a standard deviation larger than the mean, so one seed can dominate the average. The comparison between twins remains Table VII.

On the 1 h tapes the v2 tracker, which was not selected, does change the dwell labels. After burn-in on BTC, v1 issues no bull bar and the dwell label is calm on the whole series. v2 issues bull on 3.7% of post-burn bars and about 0.35 switches per year; bear and crisis dwell runs are still absent, while the argmax of p_t is bull, bear, calm and crisis on 11.6%, 11.3%, 69.7% and 7.4% of those bars. On ETH 1 h, v1 and v2 both leave about 5.4% bull and 94.7% bear. The mean matched W_2^2 on BTC 1 h falls from 4.8×10^{-5} to 2.7×10^{-5}. The trading tables above use v1 because that is what the validation score selected. Agreement between $\arg\max p_t$ and the dwell label on the holdout is 0.82 (BTC 1 h), 0.74 (BTC 15 min), 0.52 (ETH 1 h) and 0.71 (ETH 15 min).

F. Clipping and the action head

The validation weights for the native recipe were not in the repository, so seed 42 was retrained on the validation split with that recipe and the diagnostic was run on those weights. The final-stage weights and the rejected entropy-0.05 run were not used. For each of the eight twins the raw Gaussian mean was recorded on every validation bar, before the clip to $[-1, 1]$.

Table XI is that check. On every tape and both twins the raw mean lies outside $[-1, 1]$ on 95.6–99.8% of bars, and the executed position lies beyond 0.95 on 95.8–99.8% of bars. The mean absolute change in the executed position is between 0.003 and 0.062 per bar. The rule fixed before this table was read treats clipping as the cause only if at least six of the eight rows have both shares at or above 0.50 and 0.80. All eight rows clear both bars. The policy is asking for a position outside the legal interval, and the clip is what leaves it on the rail.

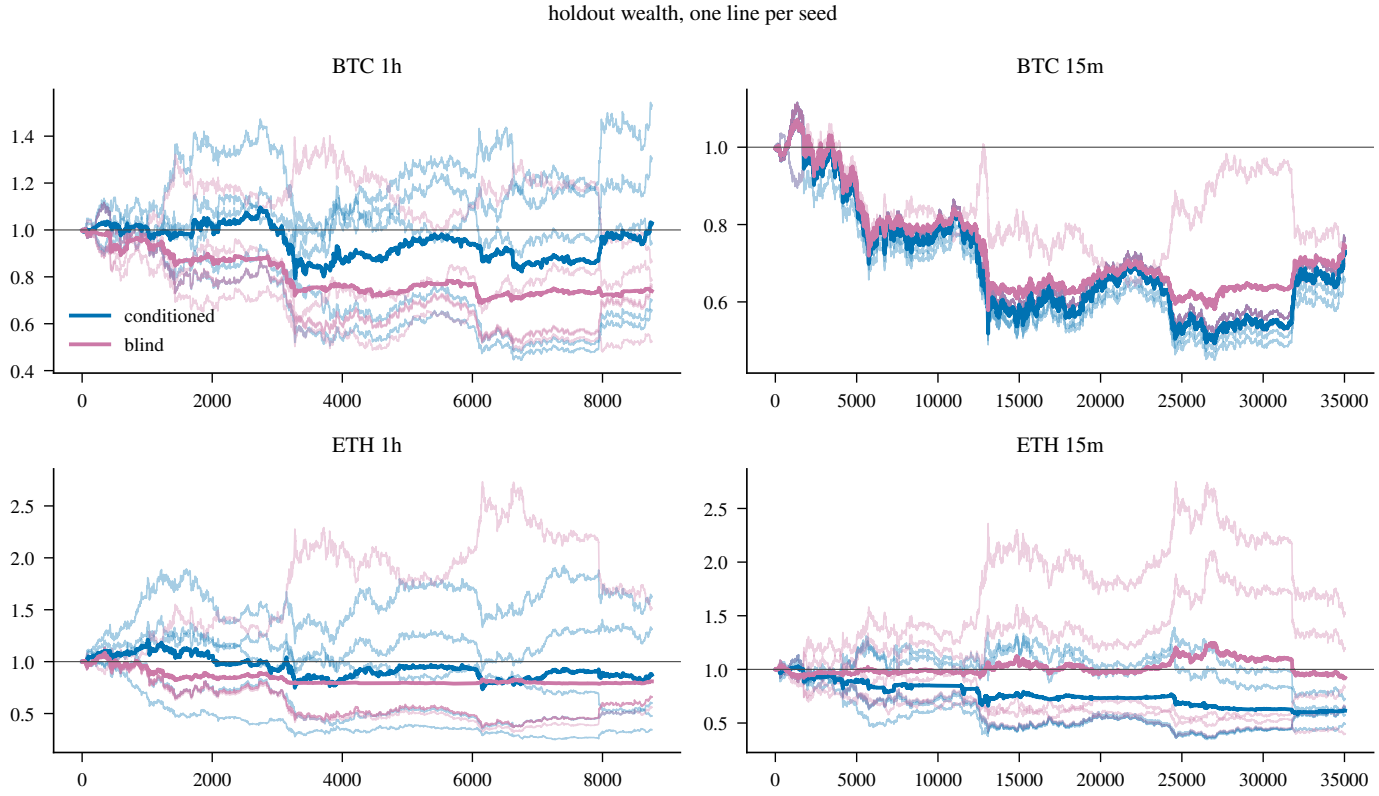


Fig. 10. Holdout wealth for the five frozen seeds. Faint lines are seeds. The heavy line is the mean. Blue is the conditioned twin and pink is the blind twin. The horizontal line is a wealth of 1.

Because that check said clipping was the cause, two other action heads were screened on the validation split only, at seed 42, with the clipped Gaussian as the control. Both twins in a run used the same head. The reward, the seed and the training budget were unchanged. `tanh_env` executes $a = \tanh(u)$. `discrete5` chooses among $\{-1, -0.5, 0, 0.5, 1\}$, and the verifier scores the expected position. The pass bar was the one already used for the native grid: both twins with position standard deviation above 0.05, both with a share of bars at $|a| > 0.95$ below 0.80, and ablation on at least 10% of bars, on all four tapes. Returns were not an input.

No head passed. The clipped Gaussian and the tanh execution stayed above the 0.80 rail share on every tape. Under tanh the BTC conditioned twins are essentially constant (position standard deviation about 0, ablation 0). The five-point head passes BTC 15 min, ETH 1 h and ETH 15 min. It fails BTC 1 h because the conditioned position is constant at 0 (standard deviation 0, ablation 0). The frozen choice is that none of the three heads is selected. Holdout and live were not scored.

VII. Limitations and Future Work

Within-window smoothing. Each HMM is fitted only on rows before its window, but decoding uses Viterbi and forward–backward over the whole 42-bar window. The probability at a bar therefore conditions on up to 41 later bars of the same window, and the dwell merge similarly needs a run to finish before it can be judged. Labels are never revised after their window closes, but they are not strictly filtered. A forward-only (filtered) posterior and an online dwell rule would remove this leakage. Until then, the per-regime results should be read with this caveat. The native tapes use the same decoder inside each weekly refit window.

Template starvation. Non-injective matching with a frozen unmatched template removes the bull regime on the four-hour tracker (Table III). One-to-one matching is not a complete fix either (Sec. VI-A). Template birth and death, and a longer or multi-cycle burn-in, are natural next steps. Tracker v2 was scored on the native validation split and was not selected. On the five-seed four-hour repetition it loses on every seed (Section VI-D).

Counterfactual design. One-hot interventions test the four-hour policy off its training distribution, because template starvation leaves some of those inputs unseen. Verifier v2 draws counterfactual regime vectors from the training distribution and rejects a flat sensitivity curve. On the five-seed twins that rate is at or below the legacy rate, and both are small (Fig. 12). The agents were not trained with a faithfulness penalty.

Statistics. The i.i.d. bootstrap ignores the serial dependence of four-hour returns. A stationary block bootstrap would give wider intervals for Tables I and IV. Tables I–V remain the published single run. Section VI-D repeats that recipe for five seeds and does not replace those tables. The repetition uses a circular block bootstrap with block length $\lceil \sqrt{n} \rceil$ and 1,000 resamples. The native comparison uses five seeds on

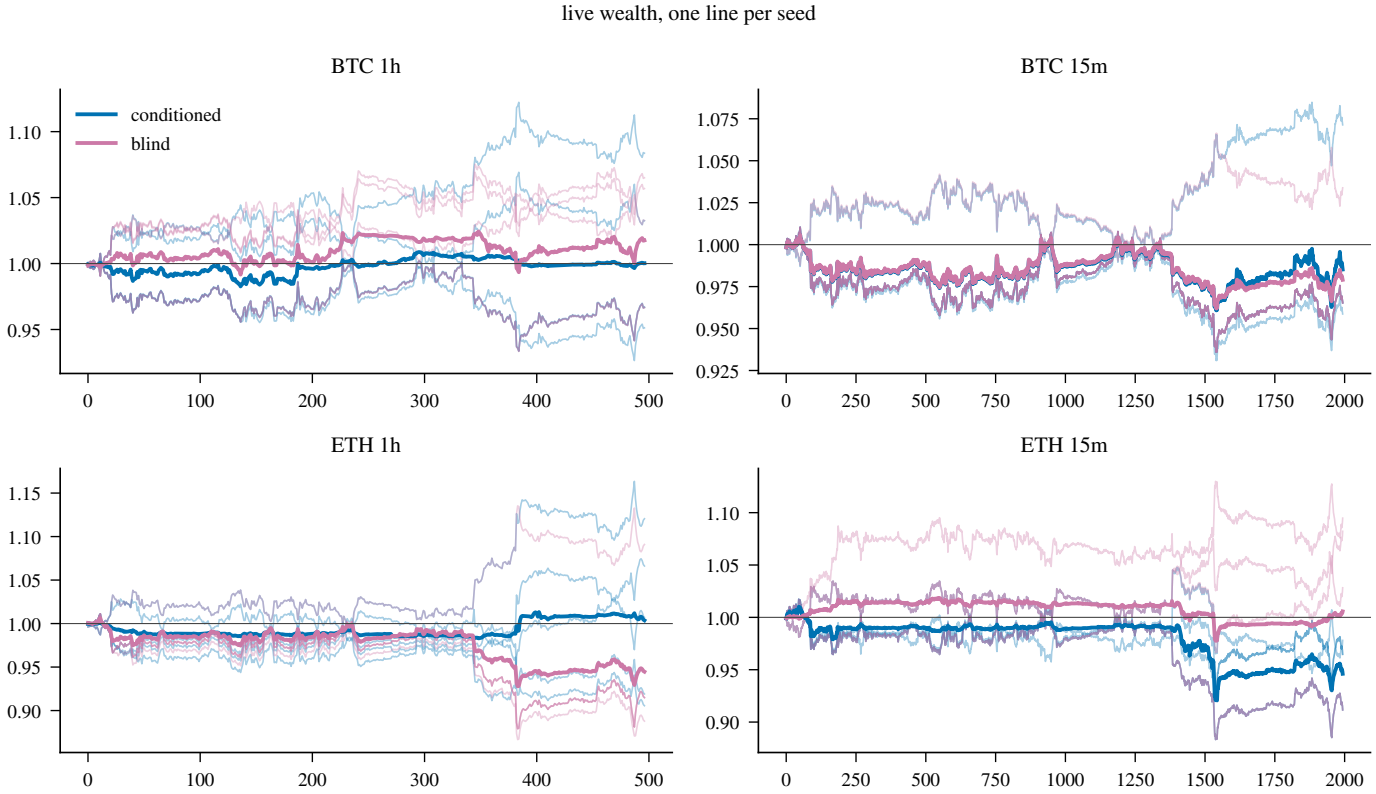


Fig. 11. Live-window wealth for the same five seeds. The scale is the three weeks after 22 Sep 2026, not the holdout year. Faint lines are seeds and the heavy line is the mean.

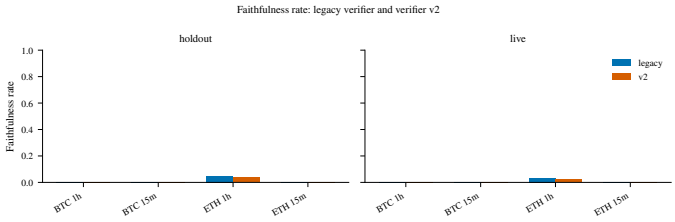


Fig. 12. Mean faithfulness rate across the five seeds. Blue is the legacy one-hot verifier. Orange is verifier v2, which uses training-set probability vectors and rejects a flat sensitivity curve. The axis runs to 1. The bars do not.

BTC/USDT and ETH/USDT. It reports the standard deviation of the five numbers and the share of seeds on which the conditioned log-return sum is larger.

Live evidence. The live window is 21 days, from 22 Sep to 13 Oct 2026, and buy-and-hold lost money on every native tape. The window can show the direction of a gap. It cannot establish one. Paper fills happen at the candle close with a flat 10-bp cost, ignoring slippage, latency and the funding cost of short positions. Across the five seeds the mean share of bars with $|a| > 0.95$ is 97–100% for every twin. The validation grid did not meet the pre-registered bar, and holdout and live were not used to choose the recipe. The console's start-up labelling is faster than the tracker, but it decodes its whole window at once, which smooths within the window over hundreds of candles. Its on-screen wealth curves are illustrative, and every live number reported here comes from the frozen study and the tracker.

Scope. The thresholds δ_{ab} and δ_{cf} and the defensive ordering ρ are design choices. Transaction costs are a flat 10 bp, and the agents were built directly on Stable-Baselines3. Ndoutoumou *et al.* place the regime signal inside the FinRL stack [1]. The four-hour sections use tracker v1 and the one-hot verifier. The native agents are not regularized.

VIII. Conclusion

We built a regime-conditioned RL trading agent with a Wasserstein identity tracker and a suite that intervenes on the regime input of every decision. On BTC/USDT four-hour data the tracker kept a stable volatility ordering across 292 causal refits and starved the bull template. The conditioned agent used those labels: removing them changed its position on 60.7% of the 11,430 out-of-fold bars, and it reduced exposure in bear and crisis segments. Only 0.1% of bars passed all three checks, against 33.5% for an archived moment-sorted labelling. The out-of-fold Sharpe ratios are -0.16 for the conditioned agent, -1.12 for the blind twin and 0.35 for buy-and-hold. These four-hour numbers were not recomputed here.

The native-twin result is five seeds of the frozen recipe on BTC/USDT and ETH/USDT, at 15 min and 1 h. The validation grid did not meet the pre-registered bar, so `configs/final.yaml` stayed the native recipe: raw reward, scale 100, entropy coefficient 0.01, uniform 2,000-

TABLE X

PER-REGIME MEANS FOR THE FIVE-SEED NATIVE TWINS, GROUPED BY $\arg\max_g p_t(g)$. ENTRIES ARE THE MEAN ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THOSE FIVE NUMBERS IN PARENTHESES. SHARE DOES NOT DEPEND ON THE SEED. REGIMES WITH ZERO SHARE ON A WINDOW ARE OMITTED. RETURNS ARE NET OF TURNOVER, IN PERCENT.

Tape	Window	Regime	Share	Cond. pos.	Blind pos.	Cond. ret. (%)	Blind ret. (%)
BTC 1 h	holdout	bear	12.2%	+0.21 (0.95)	+0.36 (0.76)	−4.3 (43.9)	−10.0 (38.1)
BTC 1 h	holdout	calm	81.9%	+0.62 (0.62)	+0.36 (0.80)	+23.8 (28.1)	+4.8 (36.1)
BTC 1 h	holdout	crisis	5.9%	+0.56 (0.29)	+0.42 (0.72)	−7.0 (20.9)	−13.2 (4.8)
BTC 1 h	live	bear	7.0%	−0.20 (1.10)	+0.60 (0.50)	−0.2 (0.5)	0.0 (0.9)
BTC 1 h	live	calm	93.0%	+0.38 (0.90)	+0.38 (0.79)	+0.2 (5.7)	+1.7 (4.9)
BTC 15 min	holdout	bear	73.8%	+0.98 (0.05)	+0.65 (0.78)	−12.2 (6.6)	−9.1 (0.2)
BTC 15 min	holdout	bull	5.5%	+1.00 (0.00)	+0.71 (0.65)	−14.2 (1.6)	−12.5 (4.7)
BTC 15 min	holdout	crisis	20.7%	+0.99 (0.02)	+0.71 (0.65)	−2.9 (7.0)	−6.3 (7.1)
BTC 15 min	live	bear	89.7%	+0.70 (0.68)	+0.60 (0.89)	+1.4 (1.5)	+1.2 (1.9)
BTC 15 min	live	bull	0.2%	+0.60 (0.89)	+0.60 (0.89)	−0.9 (1.3)	−0.9 (1.3)
BTC 15 min	live	crisis	10.1%	+0.63 (0.81)	+0.60 (0.89)	−2.0 (4.9)	−2.4 (3.7)
ETH 1 h	holdout	bear	52.3%	+0.22 (1.02)	+0.60 (0.89)	−3.8 (32.3)	+18.7 (24.3)
ETH 1 h	holdout	bull	7.9%	+0.56 (0.33)	+0.60 (0.89)	−25.8 (27.4)	+10.8 (15.0)
ETH 1 h	holdout	calm	23.0%	+0.77 (0.48)	+0.60 (0.89)	+9.3 (36.2)	+20.4 (24.9)
ETH 1 h	holdout	crisis	16.8%	−0.18 (1.02)	+0.60 (0.89)	+56.0 (119.8)	−12.4 (125.2)
ETH 1 h	live	bear	86.7%	+0.20 (1.05)	+0.60 (0.89)	−0.5 (8.5)	−4.9 (7.6)
ETH 1 h	live	bull	1.0%	+0.70 (0.28)	+0.60 (0.89)	+1.1 (1.3)	+1.0 (1.4)
ETH 1 h	live	calm	2.8%	+0.77 (0.46)	+0.60 (0.89)	0.0 (0.1)	0.0 (0.0)
ETH 1 h	live	crisis	9.5%	−0.19 (1.02)	+0.56 (0.88)	−0.3 (1.7)	−1.7 (2.4)
ETH 15 min	holdout	bull	21.6%	+0.32 (0.91)	−0.02 (0.94)	+17.1 (18.8)	+6.1 (17.4)
ETH 15 min	holdout	calm	71.1%	+0.26 (1.00)	−0.12 (0.99)	−19.9 (4.3)	−11.5 (17.8)
ETH 15 min	holdout	crisis	7.3%	+0.48 (0.56)	+0.10 (0.89)	−31.2 (24.8)	+3.3 (54.2)
ETH 15 min	live	bull	7.8%	+0.45 (0.78)	+0.04 (0.93)	+1.7 (3.0)	+1.4 (2.2)
ETH 15 min	live	calm	91.3%	+0.38 (0.87)	−0.05 (0.97)	−5.3 (1.9)	+0.8 (5.7)
ETH 15 min	live	crisis	0.9%	+0.74 (0.38)	+0.53 (0.87)	−1.7 (2.4)	−1.8 (3.7)

TABLE XI

VALIDATION-SPLIT SATURATION OF THE NATIVE TWINS, SEED 42, RETRAINED BECAUSE THE VALIDATION WEIGHTS HAD NOT BEEN KEPT. SHARE OF BARS WHOSE RAW GAUSSIAN MEAN HAS ABSOLUTE VALUE ABOVE 1, SHARE WHOSE EXECUTED POSITION HAS ABSOLUTE VALUE ABOVE 0.95, AND THE MEAN ABSOLUTE CHANGE IN THE EXECUTED POSITION.

Asset	Tape	Twin	$\lvert u\rvert > 1$	$\lvert a\rvert > 0.95$	mean $\lvert\Delta a\rvert$
BTC	1 h	cond.	0.956	0.958	0.062
BTC	1 h	blind	0.989	0.990	0.006
BTC	15 min	cond.	0.998	0.998	0.013
BTC	15 min	blind	0.998	0.998	0.003
ETH	1 h	cond.	0.980	0.982	0.030
ETH	1 h	blind	0.990	0.990	0.020
ETH	15 min	cond.	0.995	0.995	0.004
ETH	15 min	blind	0.995	0.996	0.005

bar windows, tracker v1, no perturbation and no faithfulness regularizer. The baseline had the lowest mean deficit, about 0.985. Holdout and live were scored once and were not used to choose the recipe. Every twin spends 97–100% of bars at $|a| > 0.95$ on average across the five seeds. The ETH 1 h blind twin and the BTC 15-min live blind twin are flat, and the mean holdout Sharpe ratio is negative for both twins. The conditioned twin beats the blind twin on a majority of seeds in two of eight windows only, BTC 1 h holdout ($+2.8\%$ against -26.1%, beats share 0.80) and ETH 1 h live ($+0.4\%$ against -5.5%, beats share 0.80). Table VII lists the other six windows, including BTC 15 min live, where the blind position standard deviation is 0 on all five seeds, and ETH 1 h holdout, where the conditioned mean is one large path (beats share 0.20). The conditioned agent does not win overall. The agents are not regularized, and both faithfulness rates are small.

The four-hour recipe was then repeated on seeds 42–46. Tracker v1's conditioned mean return of $+57.8\%$ is seed 46 at $+515\%$, from one test fold at $+870\%$. The other four seeds lose money, and the mean Sharpe ratio is -0.27. Tracker v2 and moment-sorted names lose on average (-83.6% and -44.0%). Faithfulness stays under 3% for both trackers and is 12.5% for moment-sorted names. On the native validation twins the raw Gaussian mean is outside $[-1, 1]$ on 95.6–99.8% of bars, so the clip is what pins the executed position to the rail. Neither a tanh execution nor a five-point discrete action passed the pre-registered bar on all four tapes, and holdout and live were not scored for those heads. A faithfulness verdict has to be recomputed when the regime layer or the training recipe changes.

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